Git Commands Training Guide

Configuration Commands

git --version

Displays the installed Git version.

git config --global user.name "John Doe"

Sets the global username used in commits.

git config --global user.email "john@example.com"

Sets the global email address used in commits.

git config --list

Displays all configured Git settings.

git config user.name

Displays the currently configured username.

Repository Commands

git init

Creates a new Git repository in the current folder.

git clone

Downloads a copy of an existing repository.

Create Commands

touch file_name.txt

Creates a new text file in the current folder.

Tracking Changes

git status

Shows modified, staged, and untracked files.

git add file.txt

Stages a specific file for commit.

git add .

Stages all changed files in the current directory.

git add file1 file2

Stages multiple files.

Commit Commands

git commit -m "message

Creates a commit with a message.

git commit -am "message"

Stages tracked files and commits them.

History Commands

git log

Shows detailed commit history.

git log --oneline

Shows condensed commit history.

git show

Shows details of a specific commit.

Branch Commands

git branch

Lists local branches.

git branch feature

Creates a new branch.

git switch feature

Switches to an existing branch.

git switch -c feature

Creates and switches to a new branch.

git branch -m new-name

Renames the current branch.

git branch -d feature

Deletes a merged branch.

Merge Commands

git merge feature

Merges a branch into the current branch.

git merge --abort

Cancels an ongoing merge.

Remote Commands

git remote -v

Lists configured remote repositories.

git remote add origin

Adds a remote repository.

git remote set-url origin

Changes the remote URL.

Push and Pull

git push origin main

Uploads commits to the remote repository.

git pull origin main

Downloads and merges remote changes.

git fetch

Downloads changes without merging.

Undo Commands

git restore --staged file.txt

Removes a file from staging.

git restore file.txt

Discards local file changes.

Tag Commands

Create an annotated tag:

git tag v1.0.0

Delete annotated tag:

git tag -d v1.0.0

Push the tag to GitHub:
git push --tags

List tags:
git tag

Typical Git Workflow

Working Directory → git add → Staging Area → git commit → Local Repository → git push → Remote Repository